Oracle Utilities Customer Care and Billing v25.10HOT

Product Fix Design

Bug 38799961

Last Modified: January 12, 2026 11:01 PM
Printed On: November 4, 2009

Oracle Utilities Customer Care and Billing Product Fix Design

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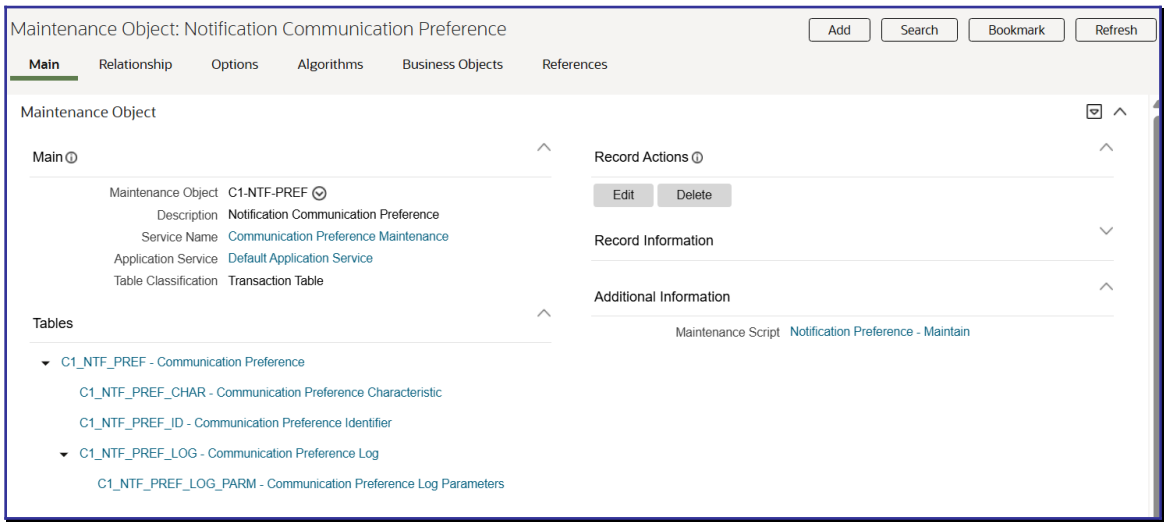
Reported Issue

Description

Unable to migrate data for maintenance object **Notification Communication Preference** ***(C1-NTF-PREF)*** specifically in tables **Communication Preference** ***(C1_NTF_PREF)*** and **Communication Preference Log** ***(C1_NTF_PREF_LOG)*** due to invalid value for the data conversion role of its tables.

Steps To Recreate

- Go To Maintenance Object
- Search for **Notification Communication Preference** ***(C1-NTF-PREF)***



Maintenance Object - C1-NTF-PREF

Tables Information

Table Role	Table	Description	Parent Constraint ID	Table Classification	System Table	Data Conversion Role
Primary	C1_NTF_PREF	Communication Preference		Transaction Table	N	Not Converted
Child	C1_NTF_PREF_CHAR	Communication Preference Characteristic	C1T003R01	Transaction Table	N	Not Converted
Child	C1_NTF_PREF_ID	Communication Preference Identifier	C1T006R01	Transaction Table	N	Not Converted
Child	C1_NTF_PREF_LOG	Communication Preference Log	C1T004R01	Transaction Table	N	Not Converted
Child	C1_NTF_PREF_LOG_PARM	Communication Preference Log Parameters	C1T005R01	Transaction Table	N	Not Converted

C1-NTF-PREF – Table Information

Expected Result: The tables that are needed to be converted has Data conversion role of **'Convert'.**

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Actual Result: The Data conversion role is **'Not Converted'.**

Solution Design

- Update the Data Conversion Role value of the **Communication Preference *(C1_NTF_PREF)*** primary table from 'Not Converted' to 'Convert (New PK)' so that the created Batch Controls can process the table properly.
- Update the Data Conversion Role value of the **Communication Preference Characteristics *(C1_NTF_PREF_CHAR)*** primary table from 'Not Converted' to 'Convert (Retain PK)' so that the created Batch Controls can process the table properly.
- Update the Data Conversion Role value of the **Communication Preference Identifier *(C1_NTF_PREF_ID)*** primary table from 'Not Converted' to 'Convert (Retain PK)' so that the created Batch Controls can process the table properly.
- Update the Data Conversion Role value of the **Communication Preference Key *(C1_NTF_PREF_K)*** primary table from 'Not Converted' to 'Convert (Retain PK)' so that the created Batch Controls can process the table properly.
- Update the Data Conversion Role value of the **Communication Preference Log *(C1_NTF_PREF_LOG)*** primary table from 'Not Converted' to 'Convert (Retain PK)' so that the created Batch Controls can process the table properly.
- Update the Data Conversion Role value of the **Communication Preference Log Parameters *(C1_NTF_PREF_LOG_PARM)*** primary table from 'Not Converted' to 'Convert (Retain PK)' so that the created Batch Controls can process the table properly.
- Create the Conversion Batch Controls by using the Maintenance Object Configuration.

Summary

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N
Does the fix involve DB Changes?	N
Is there an impact on any user interface or any other on-line behavior?	N
Is there an impact on any batch process?	**Y**
Is there an impact on CM – Custom Modification processing?	N
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

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Other Changes

Tables

Communication Preference (C1_NTF_PREF) (Existing)

Properties

Property	Value
Description	Communication Preference
Table Type	Table
Table Classification Type	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (New PK)
Maintenance Object	Notification Communication Preference
Type of Key	System- Generated

Communication Preference Log (C1_NTF_PREF_LOG) (Existing)

Properties

Property	Value
Description	Communication Preference Log
Table Type	Table
Table Classification Type	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)
Maintenance Object	Notification Communication Preference
Type of Key	System- Generated

Communication Preference Key (C1_NTF_PREF_K) (Existing)

Properties

Property	Value
Description	Communication Preference Key
Table Type	Table
Table Classification Type	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)

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Communication Preference Identifier (C1_NTF_PREF_ID) (Existing)

Properties

Property	Value
Description	Communication Preference Identifier
Table Type	Table
Table Calssification	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)
Maintenance Object	Notification Communication Preference
Type of Key	System- Generated

Communication Preference Log Parameters (C1_NTF_PREF_LOG_PARM) (Existing)

Properties

Property	Value
Description	Communication Preference Log Parameters
Table Type	Table
Table Calssification	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)
Maintenance Object	Notification Communication Preference
Type of Key	Other

Communication Preference Characteristics (C1_NTF_PREF_CHAR) (Existing)

Properties

Property	Value
Description	Communication Preference Characteristic
Table Type	Table
Table Calssification	Transaction Table
Data Conversion Role	~~Not Converted~~ Convert (Retain PK)
Maintenance Object	Notification Communication Preference
Type of Key	System- Generated

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Batch Controls

Validate C1-NTF-PREF (C1NTFPFV) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Entity Validation Template** (***F1-CVVAL***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1NTFPFV
Description	Validate C1-NTF-PREF
Template	Entity Validation Template
Selection Algorithm	Entity Validation Select Records
Processing Algorithm	Entity Validation Process Record
Detailed Description	This batch process validates entities for the specified maintenance object. By default all records are selected and each entity is fully validated for schema as well as object validation rules, at the maintenance object and business object level. The "Process Every Nth" parameter may be used to randomly select a subset of entities to validate. Processing may be further restricted to schema validation only or entity validation rules using the "Validation Mode" parameter. These options may be useful in identifying common data issues in preliminary runs. Errors encountered during validation are logged onto the conversion error table. Note that at the start of this job, all rows in the conversion error table for the process maintenance object are deleted. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	mo	Maintenance Object	Y	C1-NTF-PREF
120	validateMode	Validation Mode	N	
130	everyNth	Process Every Nth	N	
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	

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920	maxErrors	Override maximum errors	N	

Resolve XML for C1-NTF-PREF (C1NTFPFR) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **XML Resolution Template** (***F1-CVXML***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1NTFPFR
Description	Resolve XML for C1-NTF-PREF
Template	XML Resolution Template
Selection Algorithm	XML Resolution Select Records
Processing Algorithm	XML Resolution Process Record
Detailed Description	This batch process resolves foreign keys that may reside in an XML storage field on any table of the specified maintenance object. An XML storage field is either defined as CLOB or XML data type. Refer to the Process Record algorithm for more information about resolution rules. Errors are logged onto the conversion error table. Note that at the start of this job, all rows in the conversion error table for the process maintenance object are deleted. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	mo	Maintenance Object	Y	C1-NTF-PREF
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

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Generate Keys for C1_NTF_PREF (C1NTF01K) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Assign Production Keys Template** (***F1-CVASG***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1NTF01K
Description	Generate Keys for C1_NTF_PREF
Template	Assign Production Keys Template
Detailed Description	This batch process assigns new keys for the specified table. This process is not multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_NTF_PREF
120	runType	Run Type	N	
130	startRowNumber	Start from row number	N	
140	parentFieldIds	Parent Field IDs	N	
150	parentFieldDefault	Parent Field Default	N	
160	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Insert C1_NTF_PREF (C1NTF01I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1NTF01I
Description	Insert C1_NTF_PREF
Template	Insert to Production Template

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Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table. Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative messages get logged onto the batch run tree. These messages help assess the process as the batch progresses. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_NTF_PREF
120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Insert C1_NTF_PREF_CHAR (C1NTF02I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1NTF02I
Description	Insert C1_NTF_PREF_CHAR
Template	Insert to Production Template
Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table.

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	Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative messages get logged onto the batch run tree. These messages help assess the process as the batch progresses. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_NTF_PREF_CHAR
120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Insert C1_NTF_PREF_ID (C1NTF03I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1NTF03I
Description	Insert C1_NTF_PREF_ID
Template	Insert to Production Template
Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table. Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative

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	messages get logged onto the batch run tree. These messages help assess the process as the batch progresses. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_NTF_PREF_ID
120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Insert C1_NTF_PREF_LOG (C1NTF04I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1NTF4I
Description	Insert C1_NTF_PREF_LOG
Template	Insert to Production Template
Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table. Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative messages get logged onto the batch run tree. These messages help assess the process as the batch progresses.

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	This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_NTF_PREF_LOG
120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Insert C1_NTF_PREF_LOG_PARM (C1NTF05I) (New)

- After creating Batch Controls using the Maintenance Object Configuration, the **Insert to Production Template** (***F1-CVINS***) Batch Control will be duplicated for the Notification Communication Preference MO.

Properties

Property	Property Value
Batch Control	C1NTF05I
Description	Insert C1_NTF_PREF_LOG_PARM
Template	Insert to Production Template
Detailed Description	This batch process inserts converted data to production for the specified table. If the table is associated with a key table then the batch process also copies the key table. Note that because processing leverages set-based SQL the "Override number of records to commit" must always be 1. Additionally, if "trace output" is enabled, more informative messages get logged onto the batch run tree. These messages help assess the process as the batch progresses. This process is multi-threaded. Records between commits: N/A Minutes between cursor re-initiation: N/A

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Parameters

Sequence	Parameter Name	Description	Required	Parameter Value
110	tableId	Table Name	Y	C1_NTF_PREF_LOG_PARM
120	commitCount	Number of Commits	N	
130	overrideSqlTimeout	Override SQL Timeout	N	10800
900	emailMode	Email Mode	N	
910	DIST-THD-POOL	Thread Pool Name	N	
920	maxErrors	Override maximum errors	N	

Data Clean-Up / Fixes

- N/A

Recommended Test Cases

Test Case 1: Reported Issue – Conversion process for C1-NTF-PREF

- Do the prerequisite steps before running the Conversion Batch Controls
- Run the Conversion Batch Controls
  - Run the Entity Validation – **Entity Validate C1-NTF-PREF** (***C1NTFPFV***) Batch Control.
  - Run the XML Resolution - **Resolve XML for C1-NTF-PREF** (***C1NTFPFR***) Batch Control.
  - Run the Assign Production Keys - **Generate Keys for C1_NTF_PREF** (***C1NTF01K***) Batch Control.
  - Run the Added Insertion to Production Batch Controls:
    - **Insert C1_NTF_PREF** (***C1NTF01I***).
    - **Insert C1_NTF_PREF_CHAR** (***C1NTF02I***).
    - **Insert C1_NTF_PREF_ID** (***C1NTF03I***).
    - **Insert C1_NTF_PREF_LOG** (***C1NTF04I***).
    - **Insert C1_NTF_PREF_LOG_PARM** (***C1NTF05I***).
- **Expected Result:** The Batch Controls are successfully processed.

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